

Yuno: Augmenting Human Memory in Social Contexts Through Real-Time Face Recognition and Contextual Recall

Robin Sutter
University of St. Gallen
St. Gallen, Switzerland
robin.sutter2@student.unisg.ch

Benjamin Stieger
University of St. Gallen
St. Gallen, Switzerland
benjamin.stieger@student.unisg.ch

Yannik Holenstein
University of St. Gallen
St. Gallen, Switzerland
yannik.holenstein@student.unisg.ch

Yu Sun
University of St. Gallen
St. Gallen, Switzerland
yu.sun@unisg.ch

Florian Mathis
University of Applied Sciences of the
Grisons
Chur, Switzerland
University of St. Gallen
St. Gallen, Switzerland
florian.mathis@fhgr.ch

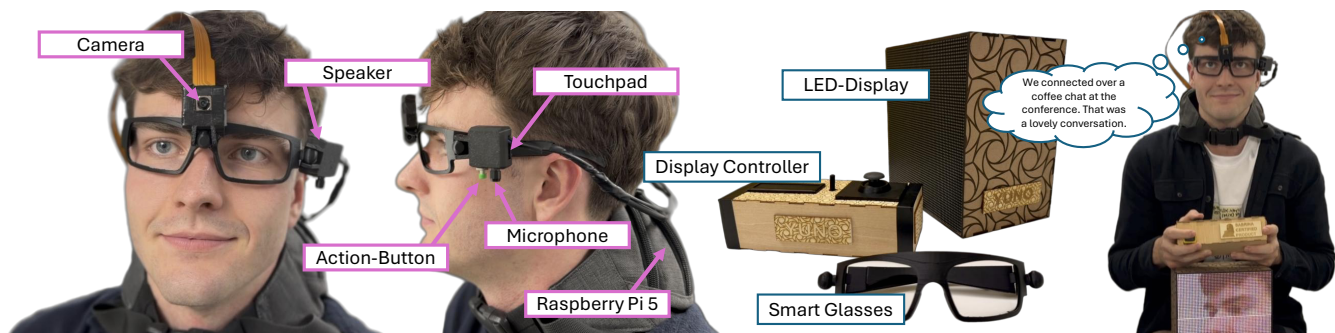


Figure 1: Yuno system overview. The system combines smart glasses for face recognition and in-situ audio reminders with a companion LED display station and controller that support gamified active recall practice.

Abstract

Remembering faces and the contexts in which people were previously encountered is cognitively demanding. In this interactivity paper, we present Yuno, an AI-powered system that augments human social memory by combining real-time face recognition with contextual recall of relationship information. Yuno integrates smart glasses that provide in-situ conversational assistance with a gamified companion that supports the practices of active recall. The smart glasses capture facial features and conversational cues during initial encounters and deliver personalized voice-based reminders during subsequent encounters. The companion 64×64 RGB LED matrix display gamifies social memory practices through AI-generated interactive quizzes, fostering engagement and relationship maintenance. Our work contributes to the HCI community by exploring the design and development of AI systems to support reconnecting

with people in socially dense environments through social memory assistance, while also pointing toward accessibility-oriented applications for individuals with face blindness (prosopagnosia) or other memory difficulties.

CCS Concepts

• **Human-centered computing** → Ubiquitous and mobile computing systems and tools.

Keywords

Face recognition, wearable computing, augmenting memory, human augmentation, AI assistant, ubiquitous computing

ACM Reference Format:

Robin Sutter, Benjamin Stieger, Yannik Holenstein, Yu Sun, and Florian Mathis. 2026. Yuno: Augmenting Human Memory in Social Contexts Through Real-Time Face Recognition and Contextual Recall. In *Designing Interactive Systems Conference (DIS Companion '26)*, June 13–17, 2026, Singapore, Singapore. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3802974.3808029>



This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License.

DIS Companion '26, Singapore, Singapore

© 2026 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2632-3/26/06

<https://doi.org/10.1145/3802974.3808029>

1 Introduction

In everyday social interactions, people often experience challenges in remembering faces, names, and associated social contexts [5],

reflecting the limits of social memory in socially dense and fast-paced environments. Despite increasingly interconnected social and professional networks, humans' cognitive capacity for face-name-context associations remains limited [10], often leading to moments of social memory overload. The retrieval of episodic memories, such as names, interests, and meeting circumstances, often fails under social pressure [20], leading to anxiety and barriers in professional networking and personal relationships. This challenge is even more profound for individuals with prosopagnosia (face blindness) or cognitive impairments due to aging or neurodegeneration [15].

Research has increasingly explored how technologies can augment human memory and cognitive capacity. Early visions such as Bush's *Memex* [3] conceptualized personal information systems designed to store, organize, and retrieve growing volumes of data. To date, many personal information systems have realized aspects of this vision by supporting the capture, storage, organization, and retrieval of personal information, and have been framed as forms of memory augmentation. These systems enable not only task reminders but also the preservation and re-experiencing of personal moments through images, videos, and contextual data. For example, everyday tools such as reminders, calendars, and alarms support cognitive offloading [17], helping users manage tasks, reduce mental workload, and improve daily functioning. Wearable recording technologies, such as SenseCam [9], capture first-person visual experiences, supporting memory recall and assisting individuals with memory impairments. With the emergence of large language models (LLMs), memory augmentation is entering a new phase. These systems enable richer semantic capture of everyday experiences in real time, allowing information to be indexed, summarized, and recalled at the level of meaning rather than raw data alone [24]. Moreover, AI-based memory systems are increasingly multimodal, enabling contextual understanding not only of spoken language, but also of faces, visual scenes, and other sensory inputs. Building on the vision of *augmenting humans* [18] and prior work on *memory augmentation* [3, 8, 12], we address the challenge of social memory by presenting Yuno: a multi-device wearable AI system designed to support remembering people and contextual details from past encounters. Yuno tackles the "who is who" and "what we discussed" problem through real-time contextual reminders, leveraging wearable computer vision [14, 22] and LLMs [2, 16]. Our system contributes a deployable, multi-device approach to supporting social memory and strengthening social connection.

2 Prototype Design and Interaction

Yuno recognizes faces, understands ongoing conversations, extracts socially relevant information, and delivers timely reminders and guidance to extend users' social memory and support real-time engagement. The system supports user interaction across four primary stages: enrollment, recognition, social support, and active recall practice. Below, we describe these four stages in detail (see also Figure 2 and Figure 3). Further details on the technical implementation are provided in the supplementary material.

2.1 Enrollment Stage

The enrollment process is designed to align with memory encoding during an initial encounter [4], capturing faces, names, locations, and other relevant information. When meeting someone new, users initiate enrollment via a single press of the Action Button on the glasses frame and then proceed with the ongoing social interaction (e.g., "Hi Sarah, nice to meet you here at the DIS Conference!"). During this interaction, the system captures audio and multiple photographs while providing subtle system feedback to indicate active capture. This design removes the need for explicit data entry and preserves the flow of social interaction, allowing enrollment to remain embedded in the ongoing social exchange. The explicit user-triggered action preserves agency and frames memory augmentation as a supportive layer within social interaction rather than a replacement for human effort.

2.2 Recognition Stage

Recognition is triggered by a double-press of the Action Button when the user encounters someone they wish to identify. A critical challenge in wearable face recognition is determining which individual in a crowded field of view is the target for interaction. Yuno employs a multi-stage logic to handle this seamlessly:

1) *Identity State Filtering*: All detected faces in the current frame are classified as either *known* or *unknown* based on their match status in the database.

2) *Automatic Novelty Detection*: During enrollment, if exactly one unknown person is found, they are automatically selected as the target, even if they are not perfectly centered. This allows the system to prioritize the "new" person in a social group, reducing the need for precise positioning.

3) *Center-Priority Heuristic*: If multiple unknown faces or only known faces are detected, the system applies a center-priority selection, choosing the face closest to the image center. This enables the user to naturally select a target by simply looking at them, aligning the system's "focus of attention" with the user's natural gaze direction.

This selection logic ensures that the system's focus matches the user's intent without requiring complex manual controls or disruptive gaze-tracking hardware, aiming to support intuitive and natural interaction.

2.3 Social Support Stage

The Social Support Stage offers two post-recognition feedback modes: *Standard Mode* and *Conversational Support Mode*.

1) *Standard Social Memory Support Mode*. In standard mode, upon successful recognition, the system delivers a personalized audio reminder through the bone-conduction speaker to provide discreet, non-intrusive feedback (e.g., "That's Sarah from the DIS Conference. You talked about her accessibility research last week"). This allows the user to re-engage confidently without interrupting the social flow. The audio output preserves ambient hearing, enabling the user to maintain awareness of their surroundings while receiving contextual information.

2) *Conversational Support Mode*. This mode extends standard recognition by providing AI-assisted conversational support. After

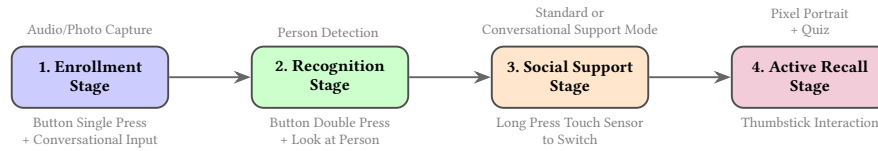


Figure 2: User interaction flow consists of four primary stages, each representing a distinct phase of the workflow, from initial enrollment to active recall practice. The on-glass interaction relies on two hardware components: an action button and a touch sensor. A single press of the button initiates the enrollment process for a new acquaintance, while a double press activates identification mode for individuals within the camera’s field of view. Following successful recognition, audio feedback is provided in the standard mode. To switch to specialized AI-supported social assistance, the user performs a long press (3–5 seconds) on the touch sensor, which activates the conversational support mode.

activation, the system performs face recognition and delivers tailored audio suggestions, such as confidence boosters, context-aware conversation openers, and socially supportive prompts for dating scenarios when relevant and appropriate to the interaction context.

2.4 Active Recall Stage: Gamified Quiz Mode

To support users’ memory consolidation of new acquaintances, the system includes a display-based review station that adopts a gamified interaction design grounded in spaced repetition principles [11]. This station supports social relationship maintenance and reorients recognition and recall practices toward more active and engaging forms of social learning. Its shared, public placement makes the process of professional networking more socially visible and participatory within collective environments.

The system transforms Yuno into a learning tool for active recall practice through a multimodal interaction design. Enrolled individuals are presented as stylized pixel-art portraits on a 64×64 RGB LED matrix, accompanied by dynamically generated multiple-choice questions based on stored episodic context (e.g., “*What research area does Sarah specialize in?*”). Additional auditory feedback, including adaptive background music that responds to interaction patterns, and text-to-speech narration, supports engagement and provides contextual information. Users interact with the quiz system via an Arduino-based controller equipped with a thumbstick and LCD display, cycling through a queue of enrolled individuals:

LEFT swipe: Users swipe left if they do not recognize the person. The system moves the current face to the back of the queue for later review, and the next person in the queue is displayed with a leftward transition animation.

RIGHT swipe: Users swipe right if they recognize the person. The system removes the current face from the queue and, if additional entries remain, displays the next person with a rightward transition. When the queue is empty, the system returns to the landing screen.

UP/DOWN: Requests context narration via text-to-speech, providing additional information about the person’s background and previous interactions.

PRESS: Displays the person’s name on the LCD display, providing a hint without removing them from the queue.

3 Situating Yuno in the Broader Sociocultural Context

While face recognition is a technically mature field [6, 19], Yuno reconfigures how social remembering is supported by integrating face recognition with conversational context capture and practice-oriented recall. LLMs enable lightweight, structured enrollment from natural interaction rather than manual data entry [2], redistributing the labor of remembering across human and computational actors. The system delivers situated conversational support that may not only facilitate re-engagement but also reshape how social continuity unfolds, embedding relational histories and shared circumstances into the conversational moment. Yuno combines on-body assistance with a shared, playful display station that externalizes memory practice as a visible social activity. The quiz mechanism, aligned with levels-of-processing principles [4], shifts from momentary assistance toward deeper semantic encoding and longer-term consolidation.

Yuno redistributes remembering across embodied interaction, networked devices, and the social spaces in which encounters unfold. It particularly targets high-load networking contexts such as conferences, where rapid introductions and brief encounters frequently lead to face–name–context breakdowns. Similar dynamics arise in hospitality and dating settings, where relational continuity and personalized recognition carry social and economic significance. In accessibility contexts, such as for people with prosopagnosia and related impairments [15, 21], memory augmentation may extend beyond convenience toward meaningful support for social participation.

Yuno is designed for consensual, assistive use: enrollment is explicit and requires informed consent. Face embeddings are processed locally and synchronized only via user-initiated backup, thereby avoiding passive surveillance.

4 Critical Reflections and Ethical Entanglements

Yuno invites reflection on how social memory is shaped by technical systems, delegated to AI, and governed through design choices that define consent, control, and relational responsibility. While such augmentation may support agency in social environments, lowering anxiety and enabling more confident participation, it also introduces a range of ethical tensions.

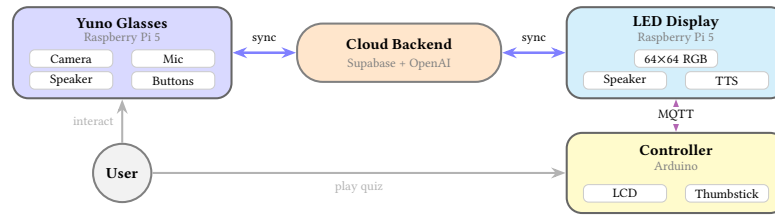


Figure 3: Yuno multi-device architecture. The wearable glasses and LED display station operate as independent Raspberry Pi 5 systems, synchronizing through Supabase cloud backend. The Arduino controller enables quiz interaction via MQTT.

One key concern is that delegating recognition to computational systems introduces surveillance-related risks. Questions emerge regarding who sets the consent rules, who controls biometric data, and how refusal is socially negotiated, as these factors determine whether users can meaningfully exercise autonomy and control over their participation. Moreover, memory technologies that enable continuous capture risk shifting from assistive recall toward ambient recording. When social encounters become permanently retrievable, intimacy and trust may be reshaped by the possibility of detailed replay and scrutiny. Cultural imaginaries such as the *Black Mirror* episode “The Entire History of You” dramatize how ubiquitous capture can transform intimacy, power, and consent by rendering past interactions endlessly examinable [1].

Face recognition further raises concerns of automated identification. Emerging AI-enabled glasses and matching systems demonstrate how personal details can be retrieved through biometric comparison with online data sources [13], creating risks of covert identification and involuntary visibility. All these intensify concerns around bystander privacy, data breaches, misrecognition, normalization of surveillance, and the social pressures that make participation in recognition systems difficult to refuse [7].

Beyond issues of surveillance and data governance, Yuno raises deeper questions about the delegation of relational labor. Remembering names and prior conversations has long been a signal of care and attentiveness. However, when such labor is partially outsourced to AI systems, what does this mean for relational responsibility? Yuno provocatively explores how AI systems can participate in the formation and maintenance of human relationships, materializing these tensions and pointing toward future research-through-design inquiries [23] into AI-mediated remembering and its implications for human connection.

5 Conclusion

This interactivity work introduces Yuno, a multi-device wearable AI system that supports social memory through real-time face recognition and contextual recall during live encounters, followed by gamified retrieval practice to support longer-term consolidation. While such systems may support agency and accessibility in socially dense environments, they also surface deeper questions about surveillance, privacy, and the delegation of relational labor.

Future work could explore real-world deployments in professional environments such as hospitality and healthcare to examine social acceptance, interactional dynamics, and long-term cognitive effects on memory consolidation, iteratively designing technology

that prioritizes social flourishing and human well-being. As AI systems increasingly participate in the constitution of social relations, Yuno invites continued inquiry into how AI-mediated remembering can be developed to support, rather than supplant, human care, responsibility, and connection in our increasingly networked social world.

Acknowledgments

Our heartfelt thanks go to Marcel Cahenzli and Eriane Breu for their tremendous support during the Hands-on HCI course at the University of St Gallen, Switzerland, and to Johannes Schöning for his continuous guidance along the way.

In this paper, Overleaf’s built-in spell checker, and the current version of ChatGPT (GPT-5.0) were used. These tools helped to fix spelling mistakes and get suggestions to improve the writing of the paper. If not noted otherwise in a specific section, these tools were not used in other forms.

References

- [1] Jesse Armstrong. 2011. *Black Mirror: “The Entire History of You”*. Television episode, Season 1, Episode 3. Directed by Brian Welsh; produced by Zeppotron for Channel 4.
- [2] Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, et al. 2020. Language Models are Few-Shot Learners. *Advances in Neural Information Processing Systems* 33 (2020), 1877–1901.
- [3] Vannevar Bush. 1945. As We May Think. *The Atlantic Monthly* 176, 1 (1945), 101–108. The foundational vision of external memory augmentation.
- [4] Fergus I.M. Craik and Robert S. Lockhart. 1972. Levels of processing: A framework for memory research. *Journal of Verbal Learning and Verbal Behavior* 11, 6 (1972), 671–684. doi:10.1016/S0022-5371(72)80001-X
- [5] G.M. Davies, J.W. Shepherd, and H.D. Ellis. 1978. Remembering Faces: Acknowledging our Limitations. *Journal of the Forensic Science Society* 18, 1 (1978), 19–24. doi:10.1016/S0015-7368(78)71178-3
- [6] Jiankang Deng, Jia Guo, Niannan Xue, and Stefanos Zafeiriou. 2019. ArcFace: Additive Angular Margin Loss for Deep Face Recognition. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 4685–4694. doi:10.1109/CVPR.2019.00482
- [7] Tamara Denning, Zakariya Dehlawi, and Tadayoshi Kohno. 2014. In Situ with Bystanders of Augmented Reality Glasses: Perspectives on Recording and Privacy-Mediating Technologies. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. ACM, 2377–2386. doi:10.1145/2556288.2557352
- [8] Jim Gemmell, Gordon Bell, Roger Lueder, Steven Drucker, and Curtis Wong. 2002. MyLifeBits: Fulfilling the Memex Vision. *ACM Multimedia* (2002), 235–238. doi:10.1145/641007.641053
- [9] Steve Hodges, Emma Berry, and Ken Woodberry. 2011. SenseCam: A wearable camera that stimulates and rehabilitates autobiographical memory. *Memory (Hove, England)* 19 (10 2011), 685–96. doi:10.1080/09658211.2011.605591
- [10] Rob Jenkins, Andrew J Dowsett, and AM Burton. 2018. How many faces do people know? *Proceedings of the Royal Society B* 285, 1888 (2018), 20181319.
- [11] Jeffrey D. Karpicke and Henry L. Roediger. 2008. The Critical Importance of Retrieval for Learning. *Science* 319, 5865 (2008), 966–968. doi:10.1126/science.1152408
- [12] Mik Lamming and Mike Flynn. 1997. Forget-me-not: Intimate Computing in Support of Human Memory. In *Proceedings of FRIEND21: International Symposium*

- on Next Generation Human Interface. Meguro Gajoen, 125–128.
- [13] Library Innovation Lab, Harvard Law School. 2025. I-XRAY: the AI glasses that reveal anyone's personal details. <https://lil.law.harvard.edu/events/i-xray-lunch/>. Lunch conversation with the student creators of the viral I-XRAY project.
 - [14] Steve Mann. 2004. Continuous Lifelong Capture of Personal Experience with EyeTap. In *Proceedings of the 1st ACM Workshop on Continuous Archival and Retrieval of Personal Experiences (CARPE'04)*. Association for Computing Machinery, 1–21. doi:10.1145/1026653.1026654
 - [15] Steve Mann, Zhiyang Pan, Yi Tao, Anqi Gao, Xingchen Tao, Danson Garcia, Dawei Shi, and Georges Kannan. 2020. Face Recognition and Rehabilitation: A Wearable Assistive and Training System for Prosopagnosia. In *2020 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*. IEEE, 731–737. doi:10.1109/SMC42975.2020.9283058
 - [16] OpenAI. 2025. GPT-5 System Card. *arXiv preprint arXiv:2601.03267* (2025).
 - [17] Evan F. Risko and Sam J. Gilbert. 2016. Cognitive Offloading. *Trends in Cognitive Sciences* 20, 9 (2016), 676–688. doi:10.1016/j.tics.2016.07.002
 - [18] Albrecht Schmidt. 2017. Augmenting Human Intellect and Amplifying Perception and Cognition. *IEEE Pervasive Computing* 16, 1 (2017), 6–10. doi:10.1109/MPRV.2017.8
 - [19] Florian Schroff, Dmitry Kalenichenko, and James Philbin. 2015. FaceNet: A Unified Embedding for Face Recognition and Clustering. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 815–823. doi:10.1109/CVPR.2015.7298682
 - [20] Bennett L. Schwartz. 2002. *Tip-of-the-Tongue States: Phenomenology, Mechanism, and Lexical Retrieval*. Lawrence Erlbaum Associates, Mahwah, NJ.
 - [21] Ben Shneiderman. 2000. Universal Usability. *Commun. ACM* 43, 5 (2000), 84–91. doi:10.1145/332833.332843
 - [22] Thad Starner, Steve Mann, Bradley Rhodes, Jeffrey Levine, Jennifer Healey, Dana Kirsch, Rosalind W. Picard, and Alex Pentland. 1997. Augmented Reality Through Wearable Computing. *Presence: Teleoperators and Virtual Environments* 6, 4 (1997), 386–398. doi:10.1162/pres.1997.6.4.386
 - [23] John Zimmerman, Jodi Forlizzi, and Shelley Evenson. 2007. Research Through Design as a Method for Interaction Design Research in HCI. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. ACM, 493–502. doi:10.1145/1240624.1240704
 - [24] Wazeer Deen Zulfikar, Samantha Chan, and Pattie Maes. 2024. Memoro: Using Large Language Models to Realize a Concise Interface for Real-Time Memory Augmentation. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (*CHI '24*). Association for Computing Machinery, New York, NY, USA, Article 450, 18 pages. doi:10.1145/3613904.3642450